

Deep Hedging

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ANALYTICAL ABSTRACT

The paper asks how a portfolio of derivatives should be hedged when trading is costly, constrained or moves prices, so that the classical complete-market delta hedge is no longer optimal. The authors define the hedger's problem as the minimisation of a convex risk measure of terminal wealth over trading strategies that are themselves neural networks, taking as inputs market prices, signals and the previous position. They show that, on a finite probability space, the network-constrained optimum converges to the unrestricted optimum as network complexity grows, and that the resulting hedging price is itself a convex risk measure and, for entropic risk, equals the exponential-utility indifference price. Numerical work in Heston models recovers model deltas without frictions, produces risk-aversion-dependent strategies under conditional value-at-risk, reproduces the epsilon to the two-thirds scaling of the price correction under proportional costs, and trains a ten-instrument hedge in a few hours. The review reconstructs the formal argument, tests the reported numbers against an independent simulation, and identifies the principal weakness: the evidence is entirely synthetic, so the claimed independence from market dynamics is untested.

1 Introduction

1.1 Scope, access and conventions

PAPER Deep Hedging proposes a data-driven solution to the hedging of derivatives books under market frictions. The authors are Buehler, Gonon, Teichmann and Wood; the reviewed text is the first arXiv version of February 2018.

CRITIQUE Access note. The sandbox in which this review was prepared could not download the PDF file itself, because the network policy blocks the arXiv host. The text was therefore read through a document-conversion service that returned the paper section by section (introduction, framework, theory, numerical experiments, references). Equations, parameter values and numbers quoted below are those returned by that reading. Where a returned formula looked inconsistent, the formula was re-derived and the discrepancy is stated. Figures of the original could not be inspected as images; the original paper file is therefore not stored with this review, and statements about figures rely on the authors' text. Every simulation labelled as such in this review was produced independently and is not an official result. Bibliographic details of references other than the year (journal, volume, pages) are given from general knowledge and were not verified against the original paper.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

Whether a single trainable policy, a neural network that maps observable information to positions, can replace model-specific hedging recipes (Greeks, coupled partial differential equations) and still deliver the optimal hedge once frictions are present.

WHY DOES THE QUESTION MATTER?

In a frictionless complete market the hedge is the delta of the pricing model. Once costs, impact or limits are added, exact solutions exist only in low dimension (Hodges and Neuberger 1989; Whalley and Wilmott 1997). Trading desks need hedges for large multi-asset books, including over-the-counter products with no observable price.

WHAT WAS KNOWN BEFORE, AND WHAT WAS MISSING?

Utility-indifference pricing under frictions was understood asymptotically in one dimension; super-hedging is typically too expensive; reinforcement-learning work on trading had been applied to portfolio and execution problems but not to derivatives liabilities. Missing was a tractable, dimension-robust, friction-agnostic numerical method with an approximation guarantee.

WHAT DOES THE PAPER PROPOSE AND FIND?

Hedging strategies parametrised by semi-recurrent networks, trained by stochastic gradient descent on a convex risk measure of terminal wealth. It reports agreement with model deltas without costs, risk-aversion-dependent deltas under conditional value-at-risk, an empirical cost scaling close to two thirds, and feasibility at ten hedging instruments.

2 Research Problem and Hypothesis

2.1 Research question

PAPER Precisely: given a liability Z , a set of hedging instruments with prices S and a friction function C_T , can the infimum of a convex risk measure of terminal profit and loss over admissible strategies be computed, to arbitrary accuracy, by restricting strategies to neural networks, and does the resulting price behave as theory predicts in benchmark models?

2.2 Existing literature and gap

PAPER The authors position the work against four strands: super-hedging under frictions (Shreve, Soner and Cvitanić 1995; Rogers and Singh 2010), utility-indifference pricing (Hodges and Neuberger 1989; Davis, Panas and Zariphopoulou 1993; Whalley and Wilmott 1997), reinforcement-learning portfolio management (Moody and Wu 1997; Xu, Jiang and Liang 2017) and algorithmic execution (Du, Zhai and Lv 2009; Lu 2017). The stated gap is that none of these covers derivatives, and in particular over-the-counter derivatives, in a way that scales with the number of instruments.

2.3 Hypotheses implicit in the paper

- H1. A network-parametrised policy can approximate the optimal friction-free hedge (model delta) without being told the model.
- H2. The same architecture adapts to risk preferences and frictions by changing only the loss function and the cost function.
- H3. The hedging price correction under proportional costs scales as $\varepsilon^{2/3}$ also in a two-instrument stochastic-volatility market.
- H4. Computational effort scales mainly with the number of hedging instruments rather than with the size of the liability portfolio.

2.4 Contribution

CRITIQUE The distinctive contribution is the combination of an approximation theorem (Proposition 4.9) with a practical training scheme for general convex risk measures. Whether this is demonstrated

is addressed in Section 7; the theoretical part is demonstrated on a finite space, the practical part only on simulated Heston markets.

3 Theoretical Framework and Mechanism

3.1 The market with frictions

PAPER Trading dates are $0 = t_0 < t_1 < \dots < t_n = T$ on a finite probability space Ω . The information vector $I_k \in \mathbb{R}^r$ generates a filtration $\mathbb{F} = (\mathcal{F}_k)$. The d hedging instruments have mid-prices $S = (S_k)$, adapted to $\mathbb{F}$; no equivalent martingale measure is assumed. A strategy is an adapted process $\delta = (\delta_k)_{k=0, \dots, n-1}$ with $\delta_k \in H_k$, where H_k is the image of a continuous map $\mathbb{R}^{d(k+1)} \rightarrow \mathbb{R}^d$ vanishing at zero (this encodes position limits). For a liability Z , an initial premium p_0 and a cost function c_k with $c_k(0) = 0$, terminal wealth is

$$\text{PL}_T(Z, p_0, \delta) = -Z + p_0 + (\delta \cdot S)_T - C_T(\delta), \quad C_T(\delta) = \sum_{k=0}^n c_k(\delta_k - \delta_{k-1}), \quad (1)$$

Z	Liability payable at T by the hedger, an $\mathcal{F}_T$ -measurable random variable.
$(\delta \cdot S)_T$	Trading gains $\sum_{k=0}^{n-1} \delta_k \cdot (S_{k+1} - S_k)$ from holding δ_k over $[t_k, t_{k+1}]$.
$c_k(\cdot)$	Cost of changing the position by $\Delta \delta_k = \delta_k - \delta_{k-1}$ at date k , with $\delta_{-1} = \delta_n = 0$.
p_0	Cash received at inception; it enters additively, which is what makes cash-invariant pricing possible.

Equation (1) says that the hedger sells Z , collects a premium, runs a self-financing trading book and pays for every change in position. The supported cost families are proportional, $c_k(n) = \sum_i \varepsilon_{i,k} S_k^i |n^i|$, fixed, $c_k(n) = \sum_i c_{i,k} \mathbf{1}_{\{|n^i| \geq \square\}}$, and market-impact specifications in which the conditional law of S_{k+1} depends on the trade. Proportional costs are convex in $\Delta \delta$; fixed costs are not, a fact that matters for Proposition 3.2 below.

3.2 Convex risk measures and the hedging price

PAPER A map ρ from random variables to $\mathbb{R}$ is a convex risk measure if it is monotone decreasing ($X_1 \geq X_2 \Rightarrow \rho(X_1) \leq \rho(X_2)$), convex, and cash-invariant:

$$\rho(X + c) = \rho(X) - c, \quad c \in \mathbb{R}. \quad (2)$$

The optimal hedged risk and the indifference price are

$$\pi(X) = \inf_{\delta \in H} \rho(X + (\delta \cdot S)_T - C_T(\delta)), \quad p(Z) = \pi(-Z) - \pi(0). \quad (3)$$

PAPER Proposition 3.2: π is monotone and cash-invariant, and if C_T and H are convex then π is a convex risk measure. The proof is the standard one: for strategies δ^1, δ^2 admissible for X_1, X_2 , the convex

combination $\alpha\delta^1 + (1 - \alpha)\delta^2$ is admissible, the gains are linear and C_T convex lowers the combined cost, so ρ convexity gives $\pi(\alpha X_1 + (1 - \alpha)X_2) \leq \alpha\pi(X_1) + (1 - \alpha)\pi(X_2)$.

INFERENCE Meaning of $p(Z)$. By cash invariance, $\pi(-Z)$ is the cash that must be added so that the hedged liability has zero risk; subtracting $\pi(0)$ removes the risk-measure value of doing nothing but trading. Hence $p(Z)$ is the minimum premium at which the hedger is indifferent between selling the claim, hedged optimally, and not selling. It is a bid-ask-sensitive reservation price, not an arbitrage price: with fixed costs π may be non-convex and the price need not be monotone in position size.

PAPER Corollary 3.5 states that if $\pi(0) > -\infty$ then $\pi(X) > -\infty$ for all X . Markets with $\pi(0) = -\infty$ are called irrelevant: a profit opportunity exists that can be scaled without bound, so no price can be assigned. This replaces the no-arbitrage assumption by a weaker requirement on the hedger's own risk measure.

3.3 Two risk measures in closed form

PAPER The entropic risk measure and the optimized certainty equivalent (OCE) of a convex, non-decreasing loss ℓ are

$$\rho_{\text{ent}}(X) = \frac{1}{\lambda} \log \mathbb{E}[e^{-\lambda X}] , \quad \rho_{\ell}(X) = \inf_{w \in \mathbb{R}} \left\{ w + \mathbb{E}[\ell(-X - w)] \right\}. \quad (4)$$

PAPER Lemma 3.6: the exponential-utility indifference price $q(Z)$ equals $p(Z)$ for ρ_{ent} . The mechanism is that exponential utility $u(x) = -e^{-\lambda x}$ has constant absolute risk aversion, so wealth at inception is multiplicatively separable: $\sup_{\delta} \mathbb{E}[u(x + W_{\delta})] = -e^{-\lambda x} \inf_{\delta} \mathbb{E}[e^{-\lambda W_{\delta}}]$, and the indifference condition removes x .

EXTENSION Derivation, not reproduced in the paper, that the entropic measure is an OCE with the stated loss $\ell(x) = e^{\lambda x} - (1 + \log \lambda)/\lambda$. Set $G(w) = w + \mathbb{E}[\ell(-X - w)]$. Then $G'(w) = 1 - \lambda e^{-\lambda w} \mathbb{E}[e^{-\lambda X}]$, so the minimiser solves $e^{\lambda w^*} = \lambda \mathbb{E}[e^{-\lambda X}]$, i.e. $w^* = \lambda^{-1} \log(\lambda \mathbb{E}[e^{-\lambda X}])$. Substituting, $\mathbb{E}[e^{-\lambda X}] e^{-\lambda w^*} = 1/\lambda$ and

$$G(w^*) = w^* + \frac{1}{\lambda} - \frac{1 + \log \lambda}{\lambda} = \frac{1}{\lambda} \log(\lambda \mathbb{E}[e^{-\lambda X}]) - \frac{\log \lambda}{\lambda} = \frac{1}{\lambda} \log \mathbb{E}[e^{-\lambda X}]. \quad (5)$$

For $\ell(x) = \max(x, 0)/(1 - \alpha)$, the minimiser w^* is the α -quantile of the loss $-X$ and $G(w^*)$ is its conditional value-at-risk, the Rockafellar and Uryasev (2000) representation. This is why a single training objective $J(\theta) = w + \mathbb{E}[\ell(\cdot - w)]$ covers both measures, with w trained as one extra scalar parameter.

3.4 Network strategies and the approximation theorem

PAPER A feedforward network with bounded non-constant activation σ has the form $F = W_L \circ F_{L-1} \circ \dots \circ F_1$, $F_{\ell} = \sigma \circ W_{\ell}$, with affine W_{ℓ} . The universal approximation theorem (Hornik 1991, Theorem 4.2) says such networks approximate continuous functions on compact sets uniformly. The hedging family is semi-recurrent:

$$H^M = \left\{ \delta_k^{\theta} = F^{\theta_k}(I_0, \dots, I_k, \delta_{k-1}^{\theta}) : F^{\theta_k} \in \text{NN}_{M, r(k+1)+d, d} \right\}, \quad (6)$$

$$\pi^M(X) = \inf_{\theta \in \Theta_M} \rho(X + (H \circ \delta^\theta \cdot S)_T - C_T(H \circ \delta^\theta)). \quad (7)$$

Equation (7) replaces an infinite-dimensional optimisation over adapted processes by a finite-dimensional one over θ . Each date has its own network (parameters θ_k); the lagged position δ_{k-1} is an input, which lets the policy express path dependence through cost memory. The composition with H projects the output onto the constraint set.

PAPER Proposition 4.9: if Ω is finite with $\mathbb{P}[\{\omega_i\}] > 0$, ρ is a convex risk measure, the H_k are continuous and the cost functions are upper semicontinuous, then $\lim_{M \rightarrow \infty} \pi^M(X) = \pi(X)$.

INFERENCE Reconstruction of the proof. Fix $\square > 0$ and take $\delta^* \in H^u$ with $\rho(\cdot; \delta^*) \leq \pi(X) + \square$ (an $\square$ -optimiser exists by definition of the infimum). On a finite space every adapted δ_k is a function of finitely many points, hence continuous on a compact set. By Theorem 4.2 there are networks $\delta^n \rightarrow \delta^*$ pointwise. Since Ω is finite, convergence at every atom is convergence almost surely. Continuity of ρ on finite spaces, continuity of H_k and upper semicontinuity of c_k give $\limsup_n \rho(\cdot; \delta^n) \leq \rho(\cdot; \delta^*)$, whence $\pi^M(X) \leq \pi(X) + 2\square$ for large M . The reverse inequality $\pi^M \geq \pi$ holds because $H^M \subset H$.

CRITIQUE What the theorem does and does not say. It guarantees that the infimum over the family approaches the true infimum. It does not bound the number of units needed, does not address whether stochastic gradient descent finds the infimum (the authors concede there is no global-convergence result), and its finite-space assumption makes the statement about the empirical sample, not about the continuous-time market of the experiments.

3.5 General convex risk measures

PAPER Theorem 4.12 gives the robust representation on a finite space, $\rho(X) = \max_{Q \in \mathcal{P}} \{\mathbb{E}_Q[-X] - \alpha(Q)\}$, with minimal penalty α . Theorem 4.13 parametrises the density $dQ/d\mathbb{P} = \exp(F_\theta(\bar{I}))$ by a network under the normalisation $\mathbb{E}[\exp(F_\theta \circ \bar{I})] = 1$, which turns the problem into a min-max over two network families. The conditions are $\alpha(Q) < \infty$ for some equivalent Q , continuity of the penalty and measurability with respect to the terminal information. **CRITIQUE** This is stated as an extension, and the numerical section does not use it; the min-max training is untested in the paper.

3.6 Mechanism of the benchmark models

PAPER The numerical market is the Heston model,

$$dS_t = \sqrt{V_t} S_t dB_t, \quad dV_t = \alpha(b - V_t) dt + \sigma \sqrt{V_t} dW_t, \quad d\langle B, W \rangle_t = \rho dt, \quad (8)$$

with $S_0 = 100$, $V_0 = 0.04$, $\alpha = 1$, $b = 0.04$, $\rho = -0.7$, $\sigma = 2$. The second hedging instrument is a variance swap with price process $S_t^2 = \int_0^t V_s ds + L(t, V_t)$.

INFERENCE Derivation of L . The price must be the conditional expectation of realised variance, $S_t^2 = \mathbb{E}[\int_0^T V_s ds \mid \mathcal{F}_t]$. From (8), $\mathbb{E}_t[V_s] = b + (V_t - b)e^{-\alpha(s-t)}$, and integrating,

$$L(t, v) = b(T - t) + \frac{v - b}{\alpha} (1 - e^{-\alpha(T-t)}), \quad \square_v L(t, v) = \frac{1 - e^{-\alpha(T-t)}}{\alpha}. \quad (9)$$

The formula returned by the document-conversion reading differs in the placement of the term $(v-b)$; equation (9) is the correct conditional expectation and is the one used here.

INFERENCE Why the model delta uses two instruments. Let $u(t, s, v)$ be the call price. Its martingale part is $u_s S \sqrt{V} dB + u_v \sigma \sqrt{V} dW$, and that of the variance swap is $\square_v L \sigma \sqrt{V} dW$. Matching both Brownian components with positions δ^1 in the stock and δ^2 in the swap gives, as in the paper's equation (5.6),

$$\delta_t^1 = \square_s u(t, S_t, V_t), \quad \delta_t^2 = \frac{\square_v u(t, S_t, V_t)}{\square_v L(t, V_t)}. \quad (10)$$

The market has two independent risk sources and two instruments, hence it is complete, and this is what allows the network's output to be compared with an exact benchmark.

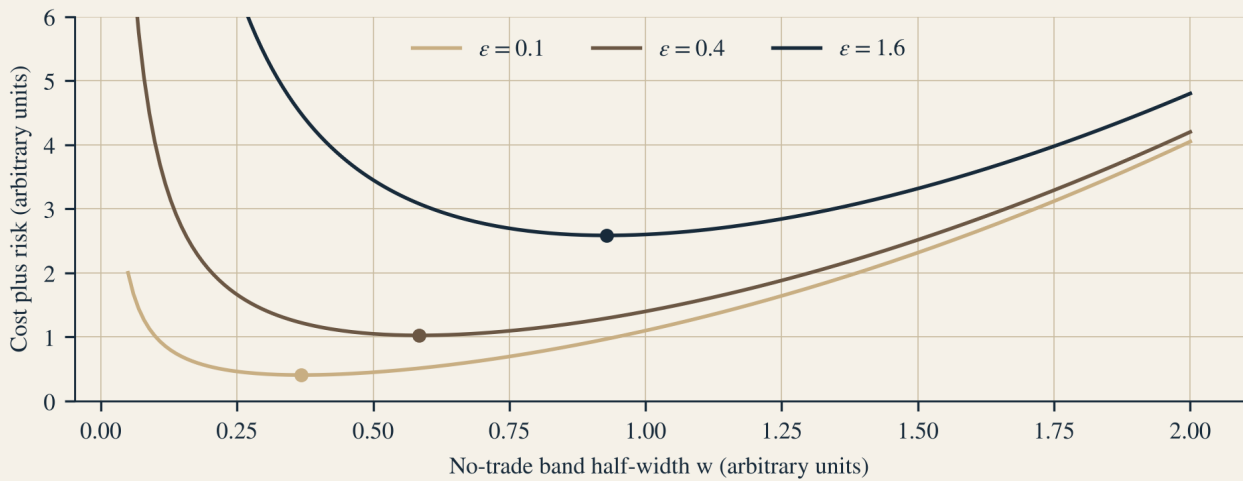
3.7 The cost scaling $\varepsilon^{2/3}$

EXTENSION Heuristic derivation (not in the paper). With proportional cost ε the hedger keeps the delta within a no-trade band of half-width w . The cost of trading scales as $A\varepsilon/w$ (smaller bands mean more trades, each of size proportional to w), and the unhedged risk charged by the risk measure scales as Bw^2 . Minimising $\Phi(w) = A\varepsilon w^{-1} + Bw^2$ gives $w^* = (A\varepsilon/2B)^{1/3}$ and

$$\Phi(w^*) = \frac{3}{2}A\varepsilon (2B/A\varepsilon)^{1/3} \propto \varepsilon^{2/3}. \quad (11)$$

This reproduces the exponent that Whalley and Wilmott (1997) derived rigorously in one dimension; it is the benchmark in Section 5.3 of the paper.

Figure 1. Cost-plus-risk trade-off in a no-trade band, for three transaction-cost levels



Source: generated for this review from the heuristic $\Phi(w) = \varepsilon/w + w^2$; dots mark the minimisers $w^* \propto \varepsilon^{1/3}$. Not taken from the paper.

4 Data and Empirical Strategy

4.1 Design

PAPER There is no empirical data. The authors simulate paths from a Heston model and treat the finite sample as the probability space Ω . Training uses approximately 10^5 to 10^6 simulated paths, and an out-of-sample test uses a further 10^6 paths. The statements about training sample size are approximate in the reading available; the exact number per experiment could not be confirmed.

4.2 Implementation

PAPER Thirty trading dates (30 days, daily rebalancing). Per-date network with three layers, dimensions $N_0 = 2d$, $N_1 = N_2 = d + 15$, $N_3 = d$, ReLU activations and batch normalisation before each activation. Adam optimiser with learning rate 0.005 and mini-batch size 256, implemented in TensorFlow. In the high-dimensional experiment the hidden width is $12 n_H$ with n_H the number of independent Heston models.

Table 1. Experimental design reported by the authors

Experiment	Market	Risk measure	Frictions
5.2 Benchmark	Heston, stock and variance swap, $d = 2$	50% and 99% CVaR	None
5.3 Scaling (a)	Black-Scholes, $\sigma = 0.2$, $d = 1$	Entropic, $\lambda = 1$	Proportional, $\varepsilon_i = 2^{5-i}$
5.3 Scaling (b)	Heston, $d = 2$	Entropic, $\lambda = 1$	Proportional, same grid
5.4 Scalability	Five independent Heston, $d = 10$	Quadratic loss $\ell(x) = x^2$, $p_0 = 5q$	None

Source: reconstructed from the authors' Section 5 as returned by the reading service. The unit of ε is not stated in the available text.

4.3 Identification

CRITIQUE Where does the identifying variation come from? Nowhere in the empirical sense: the data-generating process is known, so every claim is a claim about numerical optimisation. The variation is the random sample of paths, and the checks are against analytic benchmarks (model delta, the $2/3$ exponent, the additivity of variance). This is appropriate for a methods paper, and it also means that the paper cannot show any benefit on real prices, real cost schedules or real option books.

4.4 Data limitations

- Look-ahead and survivorship biases are absent by construction; model misspecification bias is also absent by construction, which flatters the method.

- The variance process violates the Feller condition ($2ab = 0.08$ against $\sigma^2 = 4$). In an independent Euler simulation with full truncation, about 75% of paths touched a negative variance value at least once over 30 daily steps (own simulation, 100,000 paths; an artefact of the scheme rather than a property of the exact model, but it shows how stressed the dynamics are).
- Thirty daily steps is a short horizon; the paper does not test long maturities, jumps or discontinuous liabilities.

5 Results

5.1 Result 1: recovery of the model delta without costs

PAPER Tested: whether trained networks converge to the model deltas (10). **Found:** the networks track the model deltas closely for a European call (strike $K = 100$) and a call spread ($K_1 = 100$, $K_2 = 101$) in the Heston market. For the call spread the more risk-averse strategies show flattened deltas, which the authors link to the practitioners' device of barrier shifting.

EVIDENCE The result supports H1. Its limitation is that the comparison is shown graphically; the available text gives no summary statistic such as a mean absolute delta error, so the magnitude of the discrepancy is unknown.

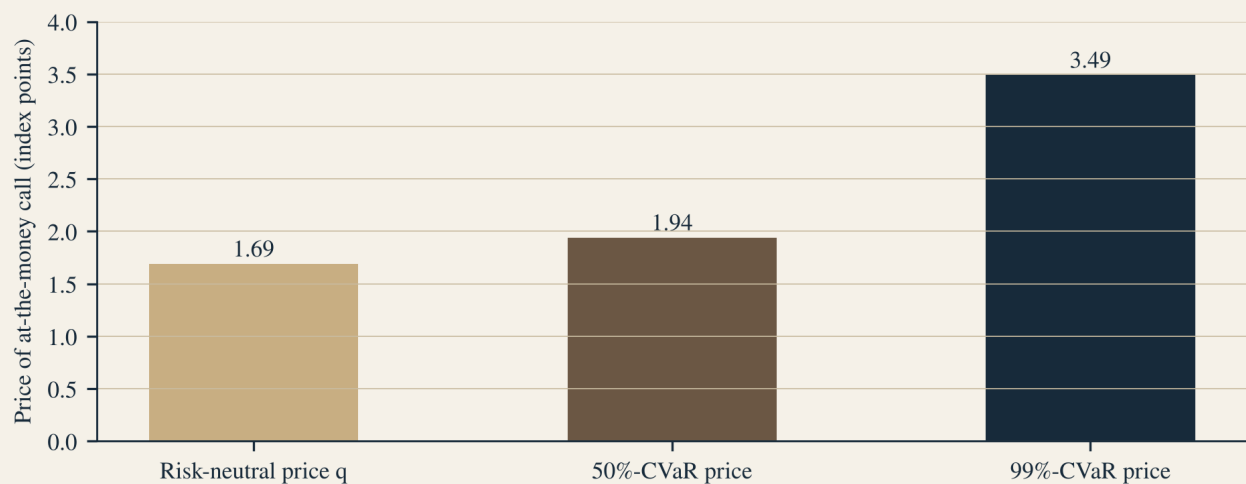
5.2 Result 2: risk-aversion-dependent prices

PAPER Tested: the price of the at-the-money call under different risk measures. **Found:** risk-neutral price $q = 1.69$; 50%-CVaR price $p_{\theta_0} = 1.94$; 99%-CVaR price $p_{\theta_0} = 3.49$.

Table 2. Prices of the at-the-money call reported in the paper (Heston, 30 days)

Risk measure	Price	Premium over q	Ratio to q
Risk-neutral, q	1.69	0.00	1.00
50%-CVaR	1.94	0.25	1.15
99%-CVaR	3.49	1.80	2.07

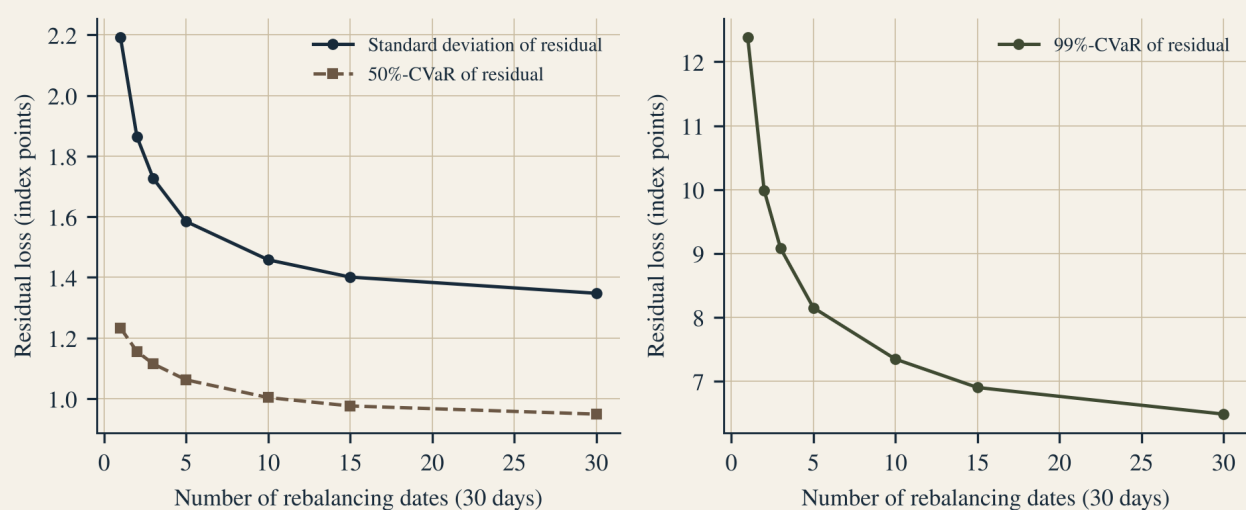
Source: values from the paper; the premium and ratio columns are computed in this review.

Figure 2. Hedging prices of the at-the-money call under three risk criteria

Source: values reported in the paper (Section 5.2). Index points, strike equal to the spot of 100.

EVIDENCE Interpretation. At a 50% CVaR level the hedger prices roughly the average loss in the worse half of outcomes, adding 0.25 index points, about 0.25 percent of the spot. At 99% the price doubles, which means that the 99% tail criterion adds about 1.8 index points to the price of the claim. Because a complete market allows perfect replication, the residual that CVaR penalises must come from the discrete-time hedge and from the network's approximation error.

CRITIQUE A consistency check. An independent Monte Carlo of the model (8) with the stated parameters, 200,000 paths, daily Euler steps and full truncation, gives a risk-neutral call price of 1.80 with standard error 0.005 (own simulation, not official). This is above the reported $q = 1.69$ by about 6.5 percent, which is many standard errors. Possible reasons are a different maturity convention, an exact rather than Euler scheme, or a parameter misread in the available text. The discrepancy is unresolved and does not affect the qualitative conclusions, but it should be checked against the original table before the 1.69 value is relied upon.

Figure 3. Residual loss of a stock-only Black-Scholes delta hedge in the Heston market, by number of rebalancing dates

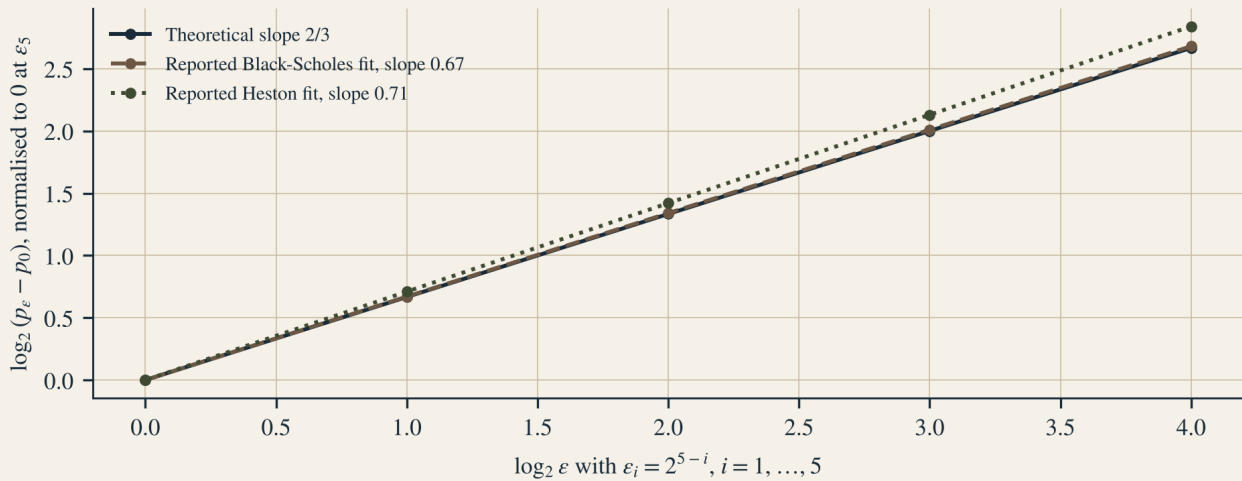
Source: generated for this review. 200,000 Euler paths, parameters of the paper, delta computed at the implied volatility 15.7% that matches the simulated call price. Not an official result. Residuals are loss minus premium; CVaR in index points.

INFERENCE The own simulation in this figure shows why the paper uses two instruments. A stock-only delta hedge with 30 dates leaves a residual standard deviation of 1.35 and a 99%-CVaR of 6.48 index points, against 2.27 and 11.46 unhedged; rebalancing more often helps little beyond ten dates because the dominant residual is the variance risk, not discretisation. The paper's prices of 1.94 and 3.49 can only be reached with the variance swap. This comparison is not between like objects (a stock-only hedge versus an optimised two-instrument hedge) and is offered only as intuition.

5.3 Result 3: transaction-cost asymptotics

PAPER Tested: whether the price correction $p_\varepsilon - p_0$ vanishes like $\varepsilon^{2/3}$. **Setup:** Black-Scholes with $\sigma = 0.2$, $s_0 = 100$, $K = s_0$, $T = 30/365$, entropic risk with $\lambda = 1$, proportional costs on the grid $\varepsilon_i = 2^{5-i}$. **Found:** a log-log regression slope of 0.67 in Black-Scholes (theory: $2/3 \approx 0.667$) and 0.71 in the Heston market with two instruments.

Figure 4. Reported log-log slopes of the price correction against the theoretical two-thirds rate



Source: slopes from the paper (0.67 and 0.71); the lines are drawn through the reported slopes and normalised to zero at the smallest cost, so they show slopes only, not the underlying data points, which were not available. Generated for this review.

EVIDENCE The Black-Scholes slope is indistinguishable from $2/3$, a useful sanity check of the whole pipeline because the exponent is known. The Heston slope of 0.71 is 6.5 percent above $2/3$; the paper presents it as consistent with the theory. **CRITIQUE** With only five points on a geometric grid and no reported standard error on the slope, a deviation of 0.04 cannot be separated from sampling noise, finite- ε corrections to the asymptotic law or optimisation error. The result is therefore supportive but not a test: no theoretical exponent exists for the Heston case in the cited literature, so the claim of a first numerical result is credible but unverified.

5.4 Result 4: scalability

PAPER Tested: ten instruments (five independent Heston models), payoff $Z = \sum_b (S_T^{2b-1} - K)^+$, quadratic loss $\ell(x) = x^2$ and $p_0 = 5q$. **Found:** test loss 0.20 and training time about 2.1 hours for one model, test loss 1.13 and about 5.75 hours for five models.

INFERENCE Calculation. For independent markets and a quadratic loss, the optimal hedged variance is additive, so the five-model loss cannot be below $5 \times 0.20 = 1.00$ if 0.20 is itself optimal. The reported ratio is $1.13/0.20 = 5.65$, which is 13 percent above the bound; training time rises by a factor $5.75/2.1 \approx 2.7$. The paper reads the ratio as near optimal. **CRITIQUE** The 13 percent gap is the cost of treating ten instruments jointly: the optimum decomposes into five two-dimensional problems, so the joint network wastes capacity learning a separable structure. This is the honest measure of the approximation error at ten dimensions, and it grows if the base loss 0.20 is not itself optimal.

5.5 Economic significance

CRITIQUE The paper contains no result expressed in trading terms: no hedging cost saving in basis points, no comparison with a practitioner benchmark such as the Leland or Whalley-Wilmott band strategies, and no analysis of the stability of the hedge to model change. The economic claim, that deep hedging outperforms the complete-market solution when frictions are present, is by construction true in a model where the objective is the same risk measure being optimised; it is not tested against a competing heuristic out of sample.

6 Robustness and Additional Tests

The paper does not contain a systematic robustness section. The checks it does contain are summarised against the corresponding threat.

Table 3. Robustness: concern, test, result and assessment

Concern	Test in paper	Result	Resolved?
Network recovers a known hedge	Compare with model delta, no costs	Close agreement, graphical	Partly
Risk-preference dependence is learned	Vary CVaR level, 50% and 99%	Prices 1.94 and 3.49 and flatter deltas	Yes for direction
Cost law is correct	Compare slope with $2/3$	0.67 and 0.71	Partly
High dimension feasible	Five independent Heston models	Loss ratio 5.65, 5.75 hours	Partly
Local minima	Not tested; stated as unresolved	None	No
Overfitting to training paths	Test on 10^6 fresh paths	Reported losses are out of sample	Yes
Misspecification of the simulator	Not tested	None	No
Seed and optimiser variability	Not reported	None	No
Architecture sensitivity	Not tested	None	No

Source: constructed from the paper's numerical section; the assessment column is the reviewer's.

CRITIQUE The most reassuring check is the use of a large fresh test sample, since the approximation theorem is stated for a finite space and a trained network can overfit the empirical measure. The least reassuring omission is the lack of repeated runs: a single training outcome for a non-convex problem is not an estimate.

7 Critical Evaluation

7.1 What the paper proves

ESTABLISHED

That, on a finite probability space with a convex risk measure, continuous constraints and upper-semicontinuous costs, the infimum over network strategies approaches the infimum over all adapted strategies (Proposition 4.9); that the optimised hedging functional is a convex risk measure under convex costs (Proposition 3.2); and that the entropic risk price equals the exponential-utility indifference price (Lemma 3.6).

SUPPORTED BUT CONDITIONAL

That a gradient-trained network recovers the Heston model delta and the $2/3$ cost exponent in Black-Scholes. The evidence depends on the particular architecture, learning rate, simulated sample size and a single reported run.

SUGGESTIVE

That the method scales to ten instruments, and that the Heston cost exponent is near $2/3$. Each rests on one number without an uncertainty estimate.

NOT ESTABLISHED

That the method is model-free in practice (every experiment uses simulated data from a known model); that it beats established frictional heuristics; that stochastic gradient descent finds the global optimum; and that the min-max scheme of Theorem 4.13 trains reliably.

7.2 Strengths

RECOGNISED BY THE AUTHORS

- Model-free and Greek-free: the policy needs only simulated or historical paths.
- Flexibility across costs, constraints and risk measures through the loss and cost functions.
- Scalability in the number of liabilities, since cost depends mainly on the number of instruments.

IDENTIFIED INDEPENDENTLY

- The convex-risk-measure formulation unifies pricing and hedging and delivers cash invariance, which makes premium a well-defined quantity even without a pricing measure. This is the conceptually strongest part of the paper.
- The semi-recurrent design feeds the previous position into the network, the correct state variable for costs with memory.
- The OCE representation makes training a single smooth expectation with a scalar auxiliary parameter, which is why mini-batch learning works.
- Validation against two analytic benchmarks (model delta; $2/3$) is the right form of test for a numerical method.

7.3 Weaknesses

CRITIQUE Each item gives the problem, why it matters, the potential bias and a possible test.

W1. SINGLE MOST IMPORTANT WEAKNESS: NO EVIDENCE BEYOND SIMULATION

Problem: every number is generated by a Heston simulator that the training also uses. **Why it matters:** the claim of independence from market dynamics is a statement about robustness to misspecification, and it is never tested. **Bias:** the networks are in-distribution by construction, so reported accuracy is an upper bound. **Test:** train on Heston paths, test on rough-volatility or jump paths, or on historical option and underlying data.

W2. NON-CONVEX OPTIMISATION WITHOUT DIAGNOSTICS

Problem: no multiple seeds, no loss curves, no check against a second optimiser. **Why it matters:** the approximation theorem is about the infimum, not the trained point. **Bias:** unknown, could be in either direction. **Test:** report the distribution of trained prices over seeds.

W3. THE THEOREM IS WEAKER THAN ITS USE

Problem: Proposition 4.9 requires finite Ω and gives no rate. **Why it matters:** the continuous-time experiments depend on discretisation and path counts. **Bias:** a sample of about 10^5 paths covers a high-dimensional path space sparsely, so the empirical measure may be poorly representative of the tails that CVaR targets. **Test:** sample-size sweep of the test loss.

W4. WEAK STATISTICAL CONTENT IN THE COST-SCALING TEST

Problem: five points, no confidence interval on the slope; 0.71 called consistent with 0.667. **Why it matters:** it is the only external validation under frictions. **Bias:** finite- ε curvature. **Test:** extend the grid and bootstrap the slope.

W5. RESERVATION PRICE VERSUS TRADABLE PRICE

Problem: the price $p(Z)$ depends on λ or the CVaR level, which the paper does not calibrate. **Why it matters:** a price moving from 1.94 to 3.49 across risk levels is not usable without a view on the risk limit. **Test:** calibrate the risk level to the observed bid-ask spread.

W6. FIXED COSTS AND NON-CONVEXITY

Problem: fixed costs are listed but Proposition 3.2 needs convex costs; no experiment uses them. **Why it matters:** lumpy costs are common in over-the-counter markets. **Test:** train under a fixed cost and check non-convexity of π in position size.

7.4 Alternative explanations

INFERENCE The flattening of deltas under high CVaR levels could also be produced by a strategy that simply holds less stock to limit tail variance, not by learned barrier shifting; the paper's interpretation is plausible but not separated from this alternative. Likewise the loss ratio of 5.65 could reflect a suboptimal one-dimensional benchmark rather than joint-training inefficiency.

8 Contradictions and Edge Cases

CRITIQUE The following edge cases were examined by reasoning and, where stated, by simulation.

LARGE RISK AVERSION

With ρ_{ent} and $\lambda \rightarrow \infty$, $\rho_{\text{ent}}(X) \rightarrow -\text{ess inf } X$, the worst-case measure, so the price tends to the super-hedging price. In the Heston market without frictions this is infinite for a call (the stock has unbounded paths), so the prices diverge; in a finite sample it is bounded by the sample maximum, which means the learned price at high λ is a statement about the simulated maximum, not about the model. The 99% CVaR price of 3.49 is already about twice the risk-neutral price; the paper does not push the level further.

VANISHING RISK AVERSION

As $\lambda \rightarrow 0$, $\rho_{\text{ent}}(X) \rightarrow -\mathbb{E}[X]$ and the hedge is no longer determined: any strategy with zero expected gain under $\mathbb{P}$ is optimal, so networks drift and the price tends to the real-world expectation of the payoff, not the risk-neutral one. Because the paper simulates under a single measure, this does not appear.

COMPLETE VERSUS INCOMPLETE MARKETS

In the benchmark (complete with two instruments) the hedging error without costs should vanish as $n \rightarrow \infty$, so the CVaR prices should converge to q . The reported premia of 0.25 and 1.80 at daily rebalancing therefore measure discretisation plus approximation error, not a property of risk aversion in the model; at finer grids they should shrink. The paper does not report this limit.

VARIANCE DYNAMICS

The parameters violate the Feller condition by a factor of 50, so the simulated variance reaches zero frequently (own simulation). Delta and variance-swap hedge ratios involve $\square_v u$ near $v = 0$, where the square-root diffusion degenerates; the benchmark model delta itself is then numerically delicate, and agreement between a network and a numerically computed benchmark does not demonstrate agreement with the exact model delta.

COST SCALING

The $\varepsilon^{2/3}$ law holds for small ε ; for large costs the optimal policy is to hold no hedge, and the price correction saturates at the unhedged value, so the exponent must fall below $2/3$ on a wide enough grid. The grid $\varepsilon_i = 2^{5-i}$ spans a factor of 16, which may straddle that regime if ε is not small.

9 Further Literature

Table 4. Literature connections

Paper	Shared idea	Relationship	Why it matters
Hodges and Neuberger (1989)	Utility-indifference price under proportional costs	Foundation; the no-trade band is the finite-dimensional analogue of the learned policy	Gives the benchmark that the network should reproduce
Davis, Panas and Zariphopoulou (1993)	Exponential-utility indifference under costs	Foundation of Lemma 3.6 and Section 5.3	Supplies the pricing definition
Whalley and Wilmott (1997)	Asymptotic cost scaling with exponent $2/3$	Provides the theoretical rate tested in Section 5.3	External validation of the numerics
Shreve, Soner and Cvitanić (1995)	Super-hedging under costs is trivial	Motivates replacing super-hedging by risk measures	Explains the choice of the objective
Föllmer and Leukert (2000)	Efficient hedging under shortfall risk	Alternative risk-constrained objective	Related criterion for partial hedging
Föllmer and Schied (2016)	Convex risk measures and robust representation	Source for Theorem 4.12	Theoretical basis of the framework
Hornik (1991)	Universal approximation	Source for Theorem 4.2	Ensures the family is dense
Moody and Wu (1997)	Learning trading policies directly	Earlier reinforcement-learning trading	Same policy-learning idea without derivatives

Paper	Shared idea	Relationship	Why it matters
Rockafellar and Uryasev (2000)	Minimisation of conditional value-at-risk	Basis of the OCE training objective for CVaR	Reviewed in this library
Gatheral and Schied (2013)	Market-impact models	Context for the impact specification	Impact is only sketched in the paper

Source: references listed in the paper; the relationships are the reviewer's classification.

INFERENCE Papers after the reviewed version are not cited here, because they could not be accessed to confirm their content. The later literature on reinforcement-learning hedging, rough-volatility hedging with neural networks and path-dependent signatures is where the unresolved points above are most likely to be addressed; the next-reading list below therefore names only works cited by the paper or already reviewed in this library.

10 Research Gaps

10.1 Paper-level gaps

- Gap: performance under misspecification. Why it matters: the headline claim is model-freedom. Existing limitation: all experiments are in-model. Way forward: train and test in different models, including rough volatility (Gatheral, Jaisson and Rosenbaum 2018, reviewed in this library).
- Gap: uncertainty quantification of trained prices. Why it matters: a price from one optimisation run is not an estimate. Existing limitation: single runs. Way forward: repeated seeds and confidence bands.
- Gap: a rate in Proposition 4.9. Why it matters: practitioners need to know how many units and paths. Existing limitation: qualitative convergence. Way forward: approximation rates under smoothness assumptions.

10.2 Literature-level gaps

- Gap: how risk-measure calibration should be linked to observed bid-ask quotes, so that reservation prices become market-comparable.
- Gap: a theory of generalisation for hedging policies trained on simulated paths and applied to real ones.

10.3 Methodological gaps

- Gradient-based training has no global guarantee; global-search or convex relaxations on small problems would calibrate the loss of accuracy.
- Costs with memory and fixed components break convexity; the corresponding theory is missing.

11 Research Opportunities

Idea 1. Deep hedging under model misspecification

RESEARCH QUESTION. How much hedging performance is lost when a network trained in a Heston simulator hedges paths from a rough-volatility or jump-diffusion market?

MOTIVATION. Tests the model-free claim, the paper's largest untested statement.

HYPOTHESIS. Out-of-model CVaR of the residual is materially worse than in-model, and worse than a robustly calibrated delta hedge.

ECONOMIC MECHANISM. Networks overfit the volatility-of-volatility and roughness of the training model.

LITERATURE CONNECTION. Gatheral, Jaisson and Rosenbaum (2018).

DATA. Simulated rough Bergomi paths and index option data with the underlying.

VARIABLES. Residual CVaR, delta error, turnover.

METHOD. Train/test cross-model grid with repeated seeds.

IDENTIFICATION STRATEGY. Controlled simulation; cross-model difference isolates misspecification loss.

EXPECTED CONTRIBUTION. A quantitative robustness map of the method.

MAIN RISK. Results may depend on architecture choice.

Idea 2. Statistical test of the cost exponent with confidence bands

RESEARCH QUESTION. Is the Heston price-correction exponent statistically different from $2/3$?

MOTIVATION. Resolves the ambiguity of the 0.71 estimate.

HYPOTHESIS. On a wider grid with repeated seeds, the exponent is statistically indistinguishable from $2/3$ for small costs and drifts lower for large costs.

ECONOMIC MECHANISM. No-trade band and the saturation regime described in Section 7.

LITERATURE CONNECTION. Whalley and Wilmott (1997).

DATA. Simulated Heston paths; ε grid over three orders of magnitude.

VARIABLES. Price correction, bootstrap slope.

METHOD. Repeated training, local regression of slope against $\log \varepsilon$.

IDENTIFICATION STRATEGY. Controlled simulation with known theory in the Black-Scholes limit.

EXPECTED CONTRIBUTION. First rigorous numerical estimate of the exponent under stochastic volatility.

MAIN RISK. Compute cost; optimisation noise comparable to the effect.

Idea 3. Calibrating the risk level to observed spreads

RESEARCH QUESTION. What CVaR level or entropic parameter makes the deep-hedging reservation price match observed option bid-ask quotes?

MOTIVATION. Turns a theoretical price into a market-comparable one.

HYPOTHESIS. Implied risk levels vary with moneyness and maturity in a systematic, term-structure-like way.

ECONOMIC MECHANISM. Dealers' balance-sheet cost of residual risk.

LITERATURE CONNECTION. Davis, Panas and Zariphopoulou (1993); Hodges and Neuberger (1989).

DATA. Intraday option quotes and underlying; cost estimates from quoted spreads.

VARIABLES. Implied CVaR level, spread, moneyness, maturity.

METHOD. Invert the price map $\alpha \mapsto p_\alpha$ by bisection over trained networks.

IDENTIFICATION STRATEGY. Cross-sectional variation in costs across strikes.

EXPECTED CONTRIBUTION. A new liquidity-adjusted implied parameter.

MAIN RISK. Quotes include non-hedging components such as inventory and information costs.

Idea 4. Deep hedging with fixed costs and non-convexity

RESEARCH QUESTION. Does the learned price become non-convex in position size under fixed transaction costs, as theory suggests?

MOTIVATION. Proposition 3.2 excludes this case; over-the-counter trading has lumpy costs.

HYPOTHESIS. Price per unit falls with size at small sizes and the optimal policy has discrete trading dates.

ECONOMIC MECHANISM. Fixed costs favour infrequent large trades.

LITERATURE CONNECTION. Hodges and Neuberger (1989); the paper's fixed-cost family.

DATA. Simulated paths.

VARIABLES. Price, position size, trade frequency.

METHOD. Smoothed indicator costs and gradient training; compare with dynamic programming in one dimension.

IDENTIFICATION STRATEGY. Dynamic-programming benchmark in $d = 1$.

EXPECTED CONTRIBUTION. Extends the framework to a case it does not cover.

MAIN RISK. Non-smooth objective hampers training.

Idea 5. Sample complexity of hedging policies

RESEARCH QUESTION. How does test loss depend on the number of training paths and units for a fixed accuracy?

MOTIVATION. Proposition 4.9 offers no rate; practitioners budget by compute.

HYPOTHESIS. Excess loss decays roughly as the inverse square root of paths until an approximation floor.

ECONOMIC MECHANISM. Statistical error of an empirical risk measure.

LITERATURE CONNECTION. Hornik (1991); the paper's own Section 4.

DATA. Simulated Heston paths at 10^3 to 10^7 sample sizes.

VARIABLES. Test loss, paths, width, depth.

METHOD. Learning-curve experiments with repeated seeds.

IDENTIFICATION STRATEGY. Controlled variation of one factor at a time.

EXPECTED CONTRIBUTION. Practical guidance and a conjectured rate.

MAIN RISK. Hard to separate optimisation from statistical error.

11.1 Ranking

Table 5. Ranking of research ideas (scores from 1 to 5, reviewer's judgement)

Idea	Novelty	Feasibility	Data	Importance	Total
1. Misspecification	4	4	4	5	17
2. Exponent test	3	5	5	3	16
3. Spread calibration	5	3	3	5	16
4. Fixed costs	4	3	5	3	15
5. Sample complexity	3	4	5	3	15

Source: reviewer's scores; no official ranking exists. Ties are broken by importance.

INFERENCE The top three are Ideas 1, 3 and 2. Idea 1 addresses the largest weakness and uses only simulation. Idea 3 has the highest potential contribution but depends on data access. Idea 2 is the cheapest and cleanest and could be completed first.

12 Learning Framework

12.1 Prerequisite concepts

Stochastic discount factors and complete markets; the Heston model; convex risk measures and cash invariance; conditional value-at-risk and the Rockafellar and Uryasev representation; exponential utility; feedforward networks, backpropagation and the universal approximation theorem; stochastic gradient descent.

12.2 Core concepts learned

Hedging as risk minimisation; the indifference price as a risk-measure difference; semi-recurrent policies; training with OCE objectives; the no-trade band logic behind $\varepsilon^{2/3}$.

12.3 Important equations

Equation (1) terminal wealth; (2) cash invariance; (3) the hedging price; (4) entropic and OCE measures; (7) network family; (10) model deltas; (11) cost scaling.

12.4 Methods learned

Parametrising a control by a recurrent network and optimising a convex risk functional by mini-batch stochastic gradients; using a scalar auxiliary parameter in an OCE; validating a numerical method against analytic benchmarks.

12.5 Questions to test understanding

COMPREHENSION

1. Why does cash invariance make $p(Z) = \pi(-Z) - \pi(0)$ independent of the hedger's initial wealth?
2. What is the role of the lagged position δ_{k-1} as a network input?
3. Why does the Heston market with a stock and a variance swap admit a unique model delta?

ANALYSIS

1. Derive that the entropic risk measure is an OCE and find its minimiser w^* .
2. Why does the heuristic $A\varepsilon/w + Bw^2$ yield the exponent $2/3$?
3. Which assumption of Proposition 4.9 fails for fixed transaction costs, and which step of the proof breaks?

RESEARCH

1. Design an experiment that falsifies the model-free claim.
2. How would you estimate the standard error of the reported price 1.94?
3. Is a 13 percent excess over additivity at $d = 10$ acceptable, and how would it scale with d ?

12.6 Papers to read next

Hodges and Neuberger (1989); Whalley and Wilmott (1997); Rockafellar and Uryasev (2000), already in this library; Gatheral, Jaisson and Rosenbaum (2018), already in this library; and the published journal version of the reviewed paper, whose changes relative to the first arXiv version are unverified here.

13 Conclusion

13.1 One-sentence thesis

Hedging under frictions can be posed as the minimisation of a convex risk measure over neural-network policies, with an approximation guarantee on finite spaces and numerical agreement with classical benchmarks in simulated Heston markets.

13.2 Three most important insights

- The convex-risk-measure formulation gives a well-defined price without a pricing measure, and reduces to exponential-utility indifference pricing for the entropic case.
- OCE objectives make risk-measure training a smooth expectation, so the same code covers CVaR and entropic risk.
- The $\varepsilon^{2/3}$ law is a cost-risk trade-off, and a network reproduces it without being told the law.

13.3 Strongest evidence

The Black-Scholes cost-scaling slope of 0.67 against the theoretical 0.667, because it is an external, theory-derived target.

13.4 Biggest weakness

All evidence is synthetic and in-model, so the model-free claim is untested.

13.5 Most important research gap

Behaviour of learned hedges under model misspecification and on real option data.

13.6 Best research opportunity

A cross-model robustness study (Idea 1), executable entirely with simulation and a known benchmark.

13.7 Three papers to read next

Whalley and Wilmott (1997); Hodges and Neuberger (1989); Gatheral, Jaisson and Rosenbaum (2018).

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